

Summary of Findings

1. Order & Sales Analysis

- Order Volume Trends: Orders peaked in certain months, with fluctuations that could be tied to seasonal patterns or marketing campaigns.
- Revenue Insights: A noticeable upward trend in revenue was observed during high-order months, with a few dips that may signal operational or payment issues.
- Order Status: Majority of orders are marked as 'completed', indicating healthy fulfillment. However, a small but consistent percentage of 'cancelled' orders might need further analysis.

2. Customer Analysis

- Repeat vs. One-Time Buyers: Over 65% of customers placed only one order, suggesting significant room for improvement in retention and loyalty strategies.
- Customer Segmentation: Repeat buyers contributed a disproportionately high share of revenue, emphasizing the importance of nurturing loyal customers.
- Monthly Customer Growth: The number of new customers increased gradually, reflecting consistent onboarding, but activation and re-engagement strategies could boost retention.

3. Payment Status Analysis

- Success Rate: ~85% of all payments were successful. This is healthy but leaves a margin (~15%) of failures worth investigating.
- Payment Failures: Failure rates spiked during specific months, likely due to operational or platform-related issues. These months should be audited for anomalies.
- Mode of Payment: Most customers preferred online/digital methods, aligning with EV-driving demographics.

4. Order Details Report

- A joined analysis of order and payment datasets revealed:
 - Correlations between order values and failure rates.
 - Customer-level granularity on order frequency, value contribution, and payment reliability.

5. Customer Retention Analysis

- Cohort-Based Heatmap: Retention dropped significantly after the first month for most customer cohorts.
- 3D Heatmap (Surface Plot): Clearly shows steep drop-offs in engagement over time, with very few cohorts maintaining activity beyond month 3.
- Insight: While initial acquisition is steady, sustained customer engagement needs attention through better onboarding, loyalty programs, or value-added services.

Recommendations for Alt Mobility

- Implement a customer reactivation strategy targeting one-time buyers.
- Investigate months with higher payment failures for systemic issues.
- Explore a loyalty program or referral incentives to boost repeat purchases.
- Consider adding automated order follow-ups and payment retry mechanisms to reduce cancellations and failures.

Thank you once again, Alt Mobility, for the opportunity to dive deep into your data!